

Padhmavathy Cebolu Srinivasan

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PROFESSIONAL SUMMARY

Data Scientist with 6 years of experience in AI-driven financial solutions. Expertise in **Python, SQL, ML, NLP, big data, and Cloud solutions**. Strong communicator with cross-functional teams, skilled in **analytical thinking, Agile collaboration, and stakeholder management**. Proven track record in building and deploying scalable models, enhancing decision-making and automation.

EDUCATION

San Jose State University

Master of Science: Data Analytics

San Jose, CA

Jan. 2023 – Dec 2024

Alagappa University

Master of Business Administration: Human Resource Management

Chennai, India

Aug. 2019 – May 2021

Sathyabama University

Bachelor of Science: Computer Science Engineering

Chennai, India

Aug. 2013 – Mar 2017

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, HTML/CSS, Bash/Linux, MongoDB

Cloud Platforms: AWS (S3, EC2, RDS), GCP (BigQuery, Compute Engine), Azure (Data Services)

Big Data: Apache Spark, Hadoop, Hive

DevOps & Tools: Jenkins, CI/CD, UDeploy, GitHub, Confluence, JIRA, Excel

Visualization & Monitoring: Tableau, Splunk

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, NLP, OpenCV, LLMs, RAG, LangChain

EXPERIENCE

Data Scientist

May. 2024 – Current

CDUS Trading LLC

Chicago, IL

- Responsible for stock market trend analysis by leveraging technical indicators, financial reports, and historical datasets, integrating key performance indicators (KPIs) for data-driven insights.
- Reduced data preprocessing time by **30%** through **optimized ETL workflows in Apache Airflow**, accelerating model training and deployment.
- Successfully developed and deployed predictive **ML models** for daily stock market forecasting, integrating risk assessment frameworks with **NLP-based sentiment analysis**. Deployment on AWS improved accessibility and reduced latency by 50%.
- Improved decision-making speed by 40% by conducting rigorous back testing, validating model performance, and documenting insights for strategic adjustments.

Business Data Analyst

June 2017 – Nov 2022

Capgemini, Synchrony Bank (Client)

Chennai, India

- Cut incident resolution time by 35%, developing **SQL-based log analysis** solutions to proactively detect and diagnose application crashes.
- Enhanced system reliability by 40%, automating cloud service restarts and auto-scaling using **Python scripts**, minimizing manual interventions.
- Actively collaborated with cross-functional teams in Scrum calls to align on business goals, track sprint progress, and enhance operational efficiency.
- Built NLP-driven log parsers to automatically extract error patterns and critical event keywords from system logs, improving root cause analysis speed and accuracy.
- Designed and maintained **Splunk and Tableau dashboards** to monitor cloud service performance, enhancing real-time visibility into system health and supporting data-driven decision-making.

PROJECTS

Docubot | *Python, RAG, LLaMA-2, ChromaDB, openai, NLP*

Developed and implemented Retrieval-Augmented Generation (RAG) using LLaMA-2 (7B), integrating ChromaDB and LangChain for optimized contextual responses. Designed an efficient pipeline leveraging quantized models and OpenAI API for enhanced retrieval and generation. [Github](#)

Traffic Management System (TMS): | *Python, ML & AI, Yolo, DeepSORT, DCRNN, ReactJS, MongoDB*

Developed CCTV-based traffic flow and accident detection models using YOLOv8 for object detection. Integrated drone-based vehicle detection and tracking using YOLOv8 + DeepSORT for real-time traffic analysis. Built a multi-cluster DCRNN model for traffic speed and flow forecasting using IoT sensor data, enabling predictive analysis. [Github-Smartcity](#)